



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECJ3483-01
WEB TECHNOLOGY

Lecturer: DR. NORIZAM BIN KATMON

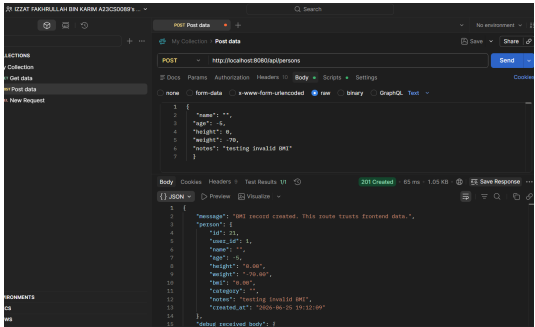
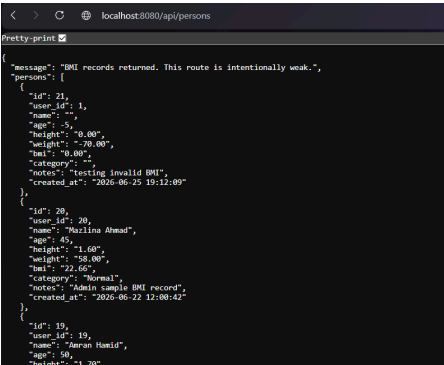
Group Project Lab Exercise

Name	Matric No
IZZAT FAKHRULLAH BIN KARIM	A23CS0089
RAMI YASSEIN ELTAYEB MOHAMED	A23CS0022
AMMAR ABDULRAHMAN ANAAM MUDHSH	A23CS0287
NOUREDIN MAMDOUH M. E. ELNAMLA	A23CS0020

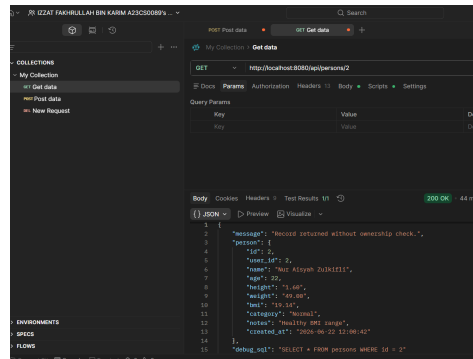
2. Security investigation summary

Test No	What I Tested	Expected Secure Behavior	Actual Result	Is It a Weakness	Why?
1	Add BMI with invalid data	Backend should reject invalid data	Backend accepts the invalid BMI data if sent through Postman/Thunder Client. Frontend validation can be bypassed.	Yes	Invalid data should not be stored because frontend validation can be bypassed.
2	Access another user's BMI record	Backend should deny access	A normal user can request another user's record using API and the backend returns it.	Yes	Users should only access records that belong to them unless they are staff/admin.
3	Access staff route as normal user	Backend should deny access	Normal user can access /api/staff/persons and view all BMI records.	Yes	Normal users should not access staff monitoring features.
4	Access admin route as normal user	Backend should deny access	Normal user can access /api/admin/users and see all users.	Yes	Admin functions must be restricted to admin role only.
5	Submit XSS payload in notes	Payload should display as text	The notes field is rendered using v-html, so payload like can execute	Yes	User-generated content should not run as JavaScript in the browser.

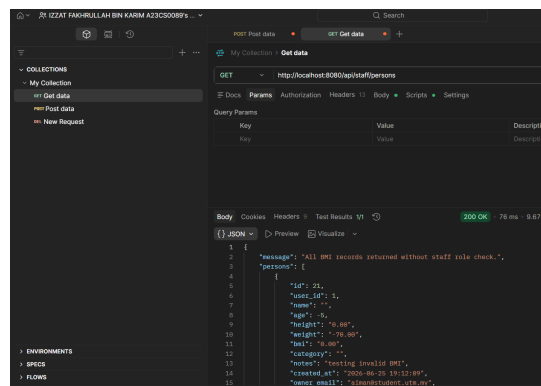
			in the browser.		
6	Check API response	Password/hash should not be visible	API responses such as /api/admin/users expose password and password_hash	Yes	API should return only necessary data and hide sensitive fields.
7	Try SQL Injection input	Input should be treated as data	Login query uses string concatenation, so SQL injection input may affect the query behavior.	Yes	SQL Injection can happen if input is directly joined into SQL queries.
8	Try to update protected fields	Backend should reject or ignore protected fields	Backend accepts update payloads containing protected fields like user_id, bmi, and category.	Yes	Users should not control fields such as user_id, role, bmi, or category.

No/Title	Screenshots
1-Add BMI with invalid data	 

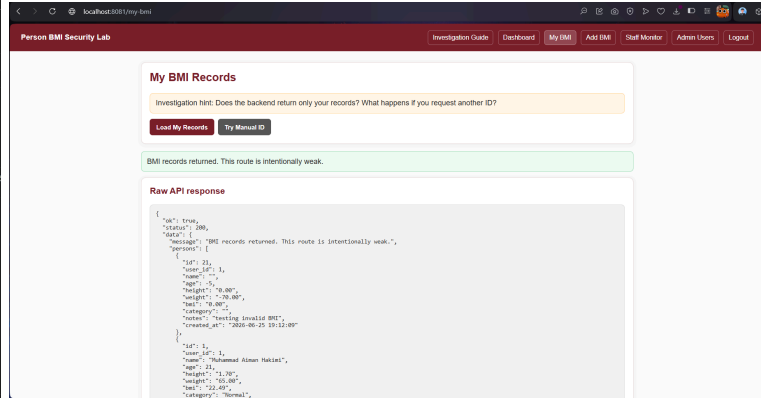
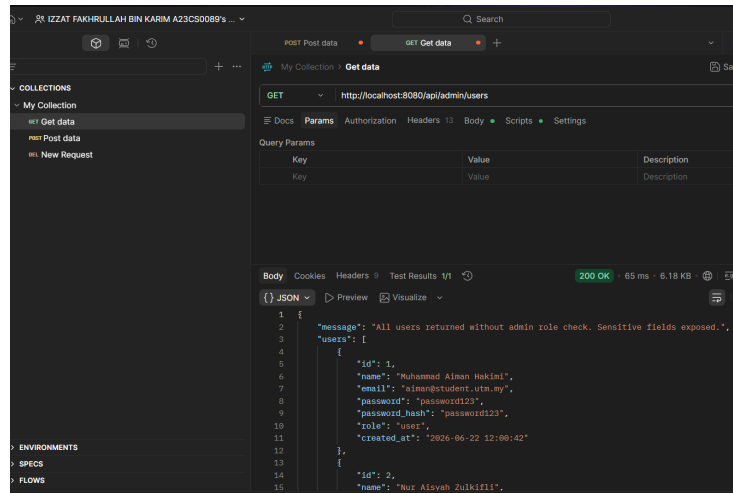
2-Access
another
user's BMI
record



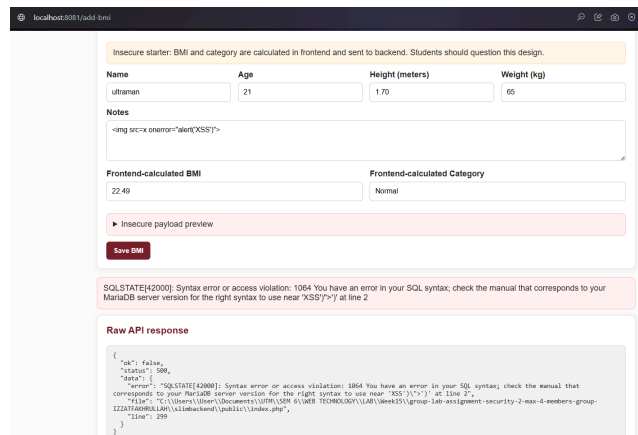
3-Access
staff route as
normal user



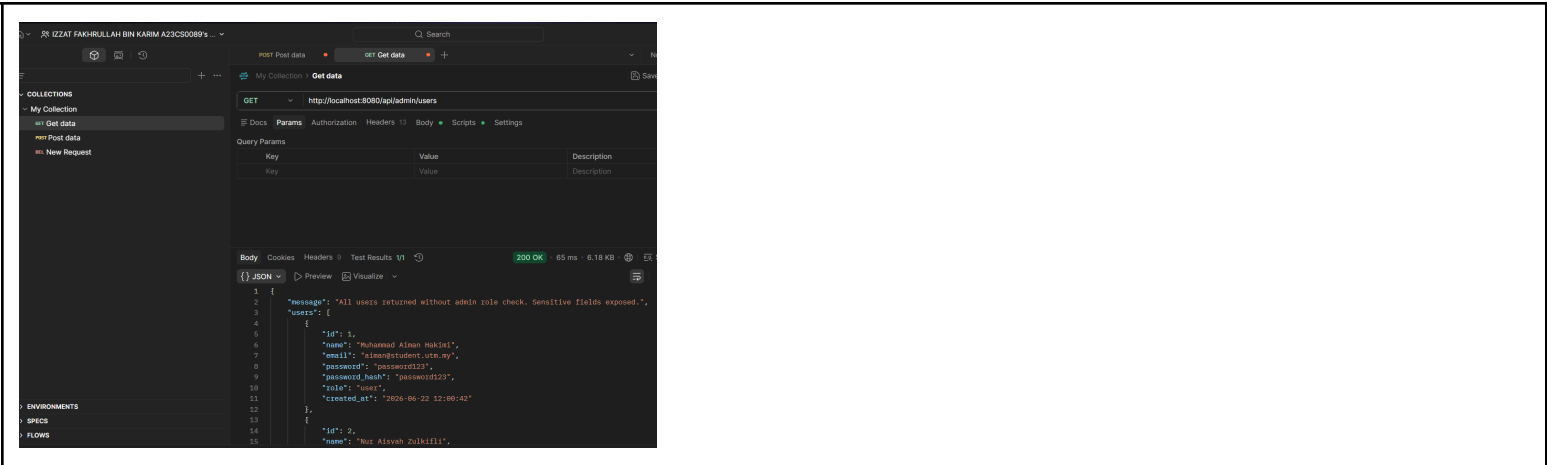
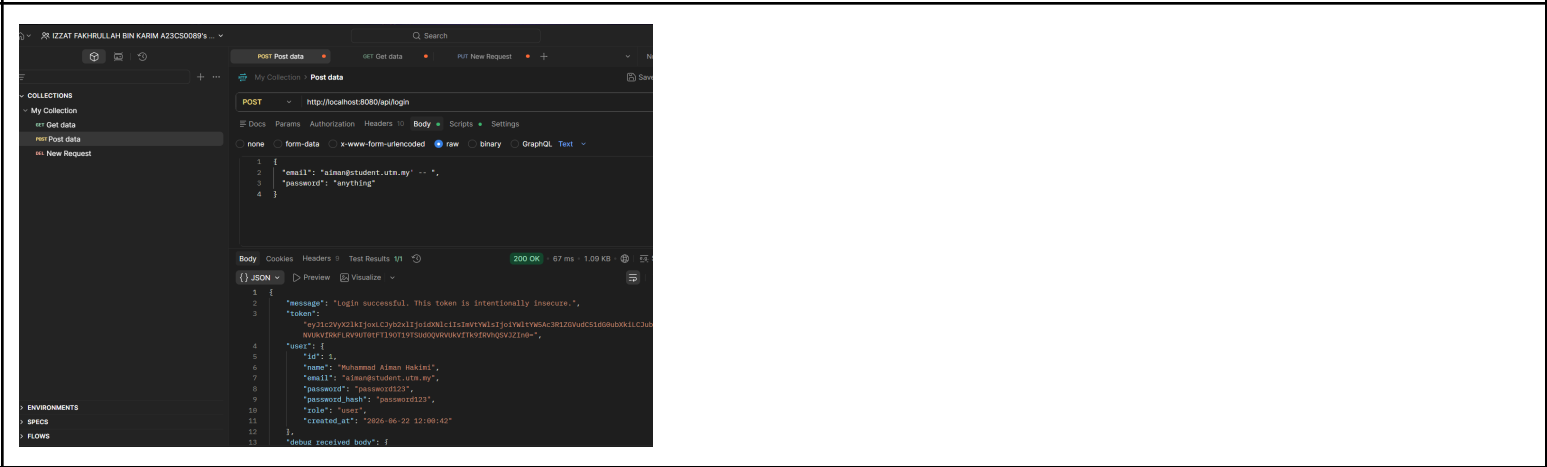
4-Access admin route as normal user



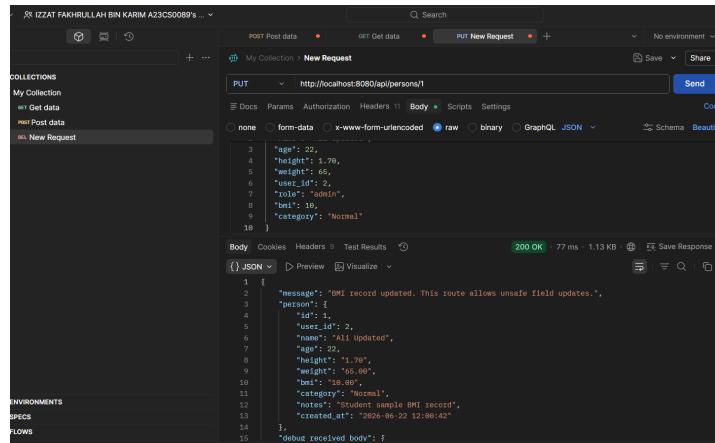
5-XSS payload in notes



6-API response exposes sensitive data

7-SQL
Injection
input

8-Update protected fields



The system contains several backend and frontend security weaknesses. The main issue is that the backend trusts user-controlled input and does not fully enforce validation, authentication, authorization, safe API responses, and safe output rendering.

